

EXP.NO: 6	Write a Servlet to demonstrate session tracking using HttpSession.
DATE:26/03/2025	

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based web application. The exercise involves implementing a simple login system where a user can log in with their username and password, and their session will be tracked throughout their interaction with the web application.

ALGORITHM:

1. Input:

- The user will enter their username and password in a login form.
- The servlet will validate the credentials and create an HTTP session for the user if authentication is successful.

2. Steps for Implementing Session Tracking with HttpSession:

1. Create the Login Form (HTML):

- Create an HTML form with two input fields for username and password.
- Provide a submit button to send the data to the servlet for authentication.

2. Create the Login Servlet:

- Write a Java servlet class to handle login requests.
- The servlet will:
 - Retrieve the username and password from the form.
 - Authenticate the user by comparing the credentials to a stored value (e.g., hardcoded or from a database).

- Create a new HTTP session for the user if the credentials are valid.
- Set session attributes such as the user's username to track the session.

3. Validate User Credentials:

- In the servlet, check if the username and password match the expected values.
- If valid, create a session and store the username in the session.
- If invalid, redirect the user back to the login page with an error message.

4. Create and Track User Session:

- Upon successful login, use the HttpSession object to store the session data.
- Use the session.setAttribute("username", username) method to save the username.
- Optionally, set session timeout using session.setMaxInactiveInterval(time_in_seconds) to manage session expiration.

5. Redirect to a Protected Page:

- After a successful login, the user will be redirected to a protected page (e.g., a dashboard or home page).
- The protected page should check if the user is authenticated by checking the session attribute.

6. Logout and Session Invalidation:

- Provide a logout feature that invalidates the session using session.invalidate() when the user logs out.
- After logout, redirect the user back to the login page.

7. Session Expiration:

- If the session expires or the user manually logs out, the session will be invalidated, and the user will be redirected to the login page.

Program:

Login.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Login</title>
  <link rel="stylesheet" href="css/login.css">

  <style>
    body{
      font-family: Arial;
      background: #dbeafe;
      text-align: center;
      margin-top: 100px;
    }

    .box{
      background: white;
      width: 350px;
      margin: auto;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 0 10px gray;
    }

    input{
      width: 90%;
      padding: 8px;
```

```
margin: 10px;  
}
```

```
button{  
padding: 10px 20px;  
background: #0d47a1;  
color: white;  
border: none;  
cursor: pointer;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class= "box">
```

```
<h2>Student Login</h2>
```

```
<form action= "login" method= "post">
```

```
<input type= "text" name= "uname" placeholder= "Enter Username"  
required>
```

```
<input type= "password" name= "pwd" placeholder= "Enter Password"  
required>
```

```
<button type= "submit">Login</button>
```

```
</form>
```

```
</div>
```

```
</body>
```

```
</html>
```

loginServlet.java:

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/login")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        String user = request.getParameter("uname");
        String pass = request.getParameter("pwd");

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        if(user.equals("admin") && pass.equals("1234")) {

            HttpSession session = request.getSession();
            session.setAttribute("username", user);

            response.sendRedirect("dashboard");
        }
        else {
            out.println("<h2 style='color:red;text-align:center;'>Invalid
Login</h2>");
        }
    }
}
```

DashboardServlet.java:

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/dashboard")
public class DashboardServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        if(session != null) {

            String user = (String) session.getAttribute("username");

            out.println("<html><body style='font-family:Arial;text-align:center;margin-top:100px;'>");

            out.println("<h1>Student Information Management System</h1>");
            out.println("<h2>Welcome, " + user + "</h2>");
            out.println("<p>Session Tracking Successful</p>");
            out.println("<a href='logout'>Logout</a>");

            out.println("</body> </html>");
        }
        else {
            response.sendRedirect("login.html");
        }
    }
}
```

LogoutServlet.java:

```
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/logout")
public class LogoutServlet extends HttpServlet {

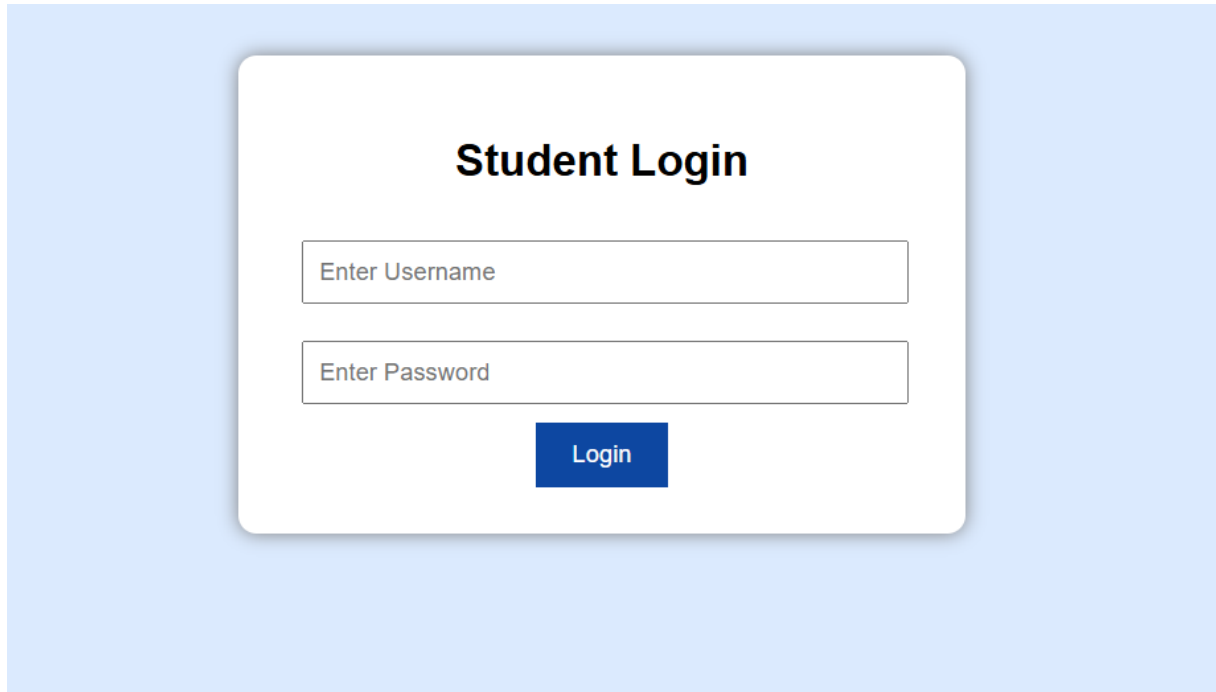
    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if(session != null) {
            session.invalidate();
        }

        response.sendRedirect("login.html");
    }
}
```

Output:

A screenshot of a web application's login page. The page has a light blue background. In the center, there is a white rounded rectangle with a subtle drop shadow. Inside this rectangle, the text "Student Login" is centered at the top in a bold, black font. Below the title, there are two input fields: the first is labeled "Enter Username" and the second is labeled "Enter Password". Both labels are in a light gray font. Below the password field, there is a blue button with the word "Login" in white text.

Student Login

Enter Username

Enter Password

Login

Student Information Management System

Welcome, admin

Session Tracking Successful

[Logout](#)

RESULT:

A simple login system is created where the user's session is tracked by a servlet to demonstrate session tracking using HttpSession.